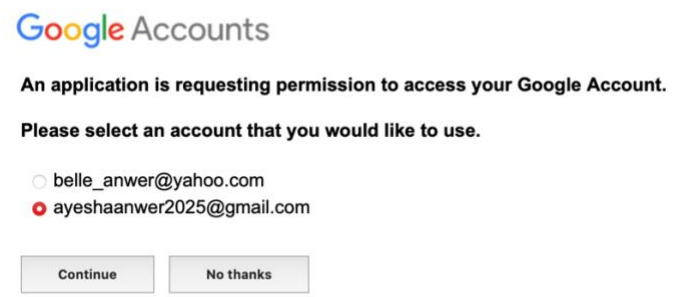
Phase 1: GEE Project Set Up

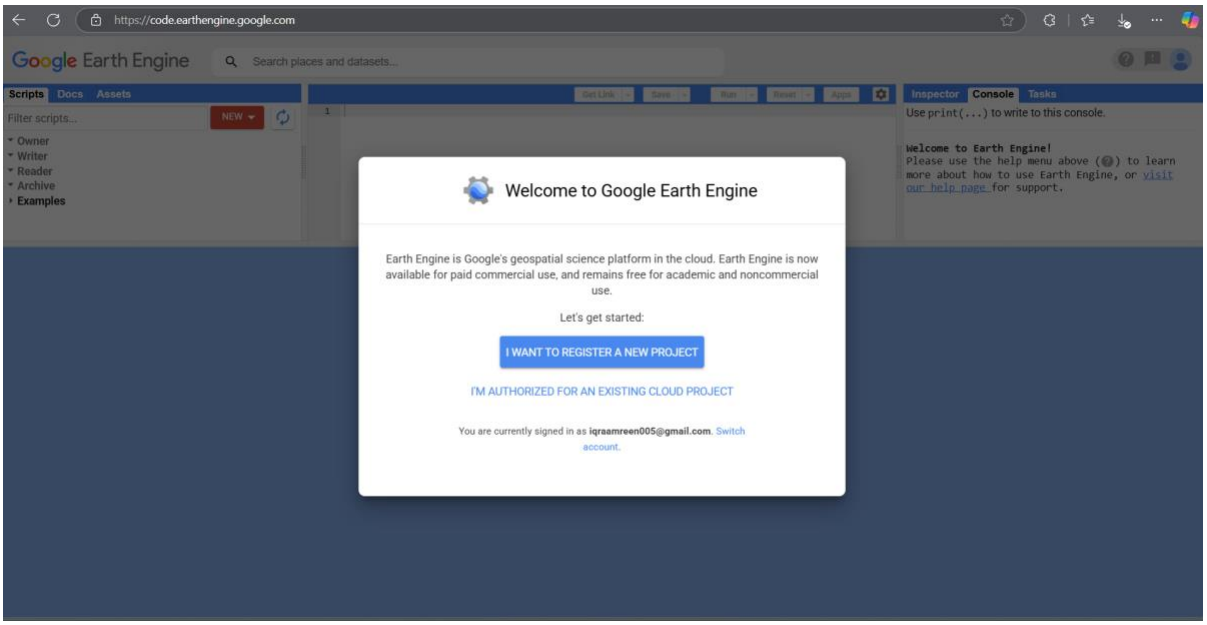
1. Start at the link below:

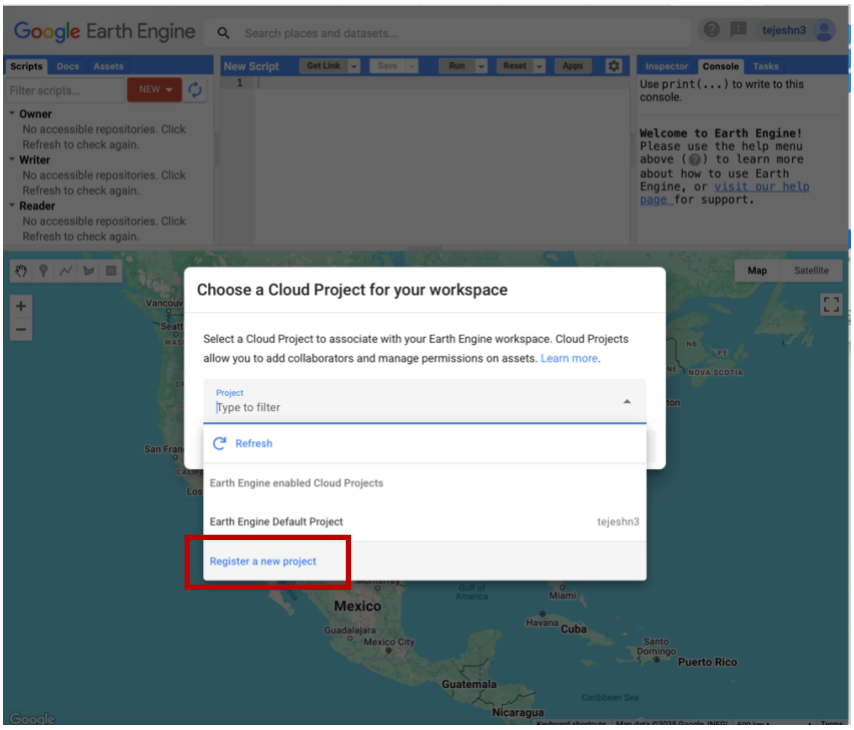
Visit http://code.earthengine.google.com

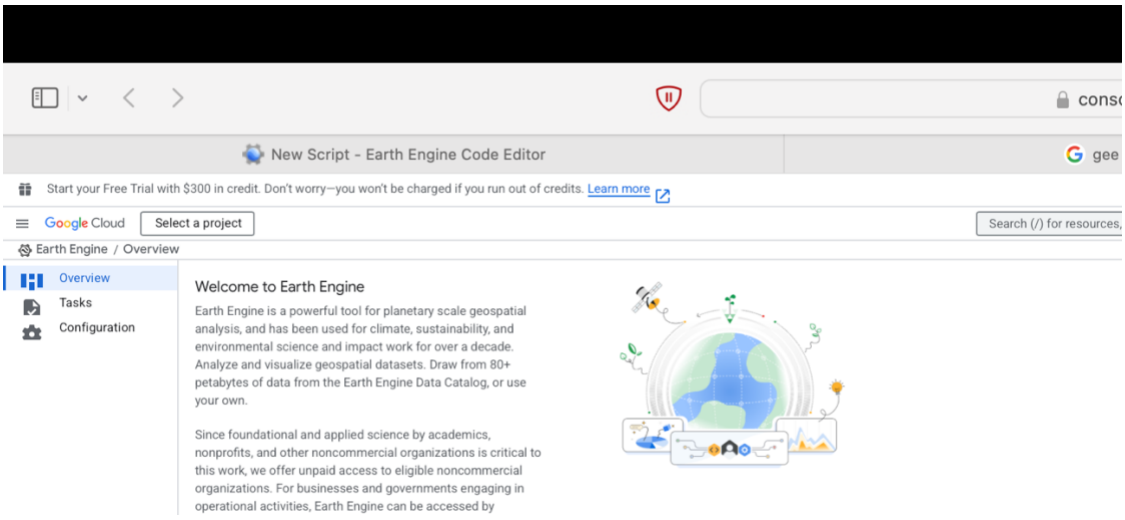


Select the google account you want to set up GEE code editor with. Make sure to use the same google account for google colab code editor.

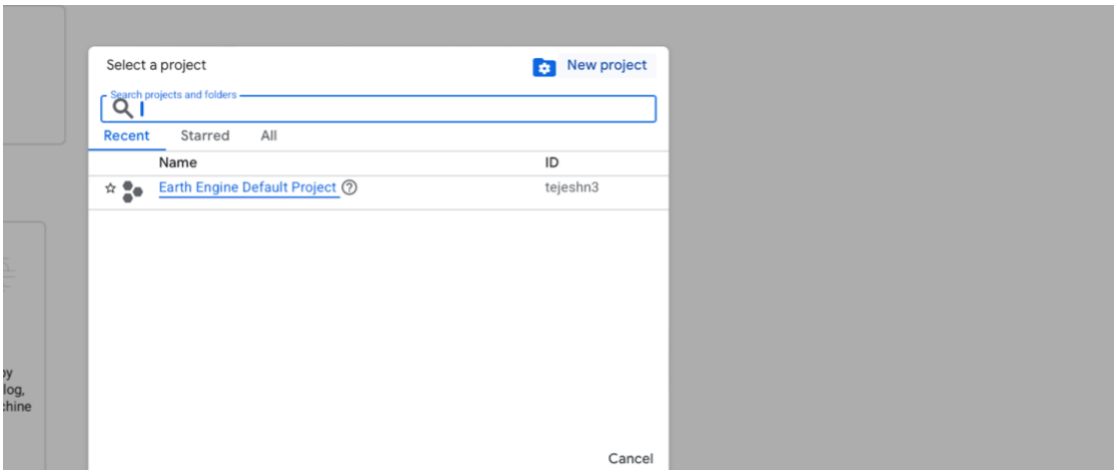
2. Register a new project.



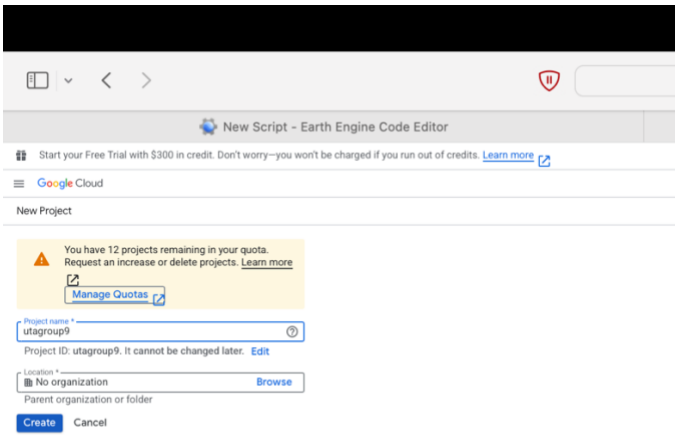


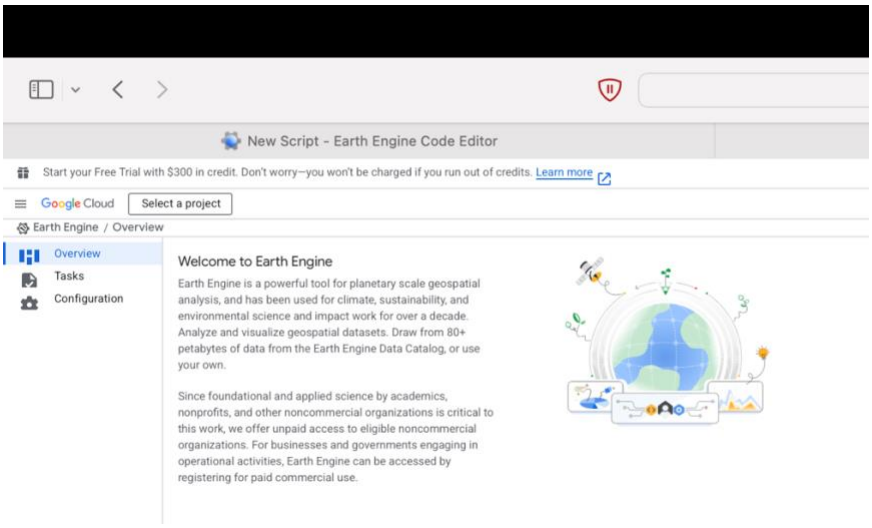


Select “Select a project”

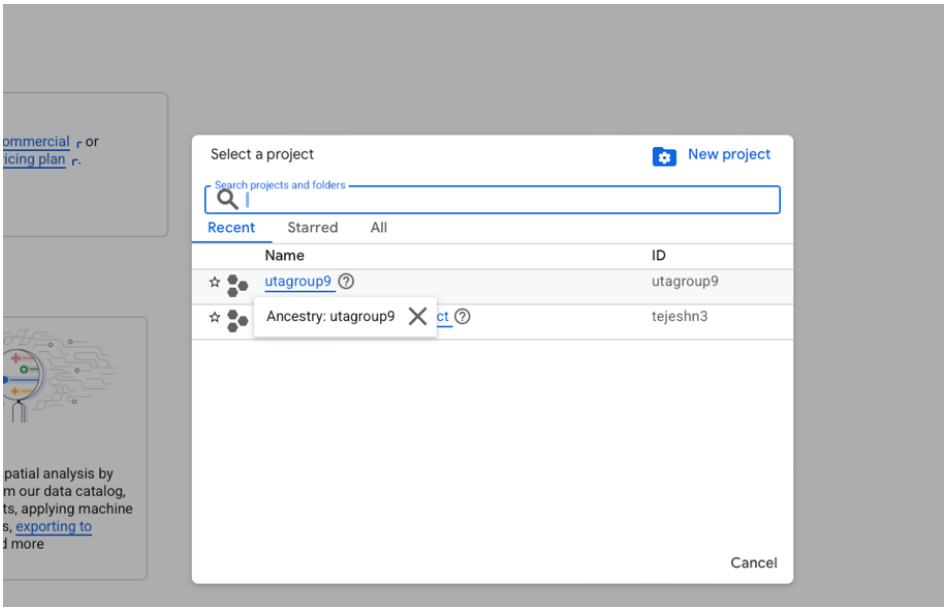


Click on “New project”



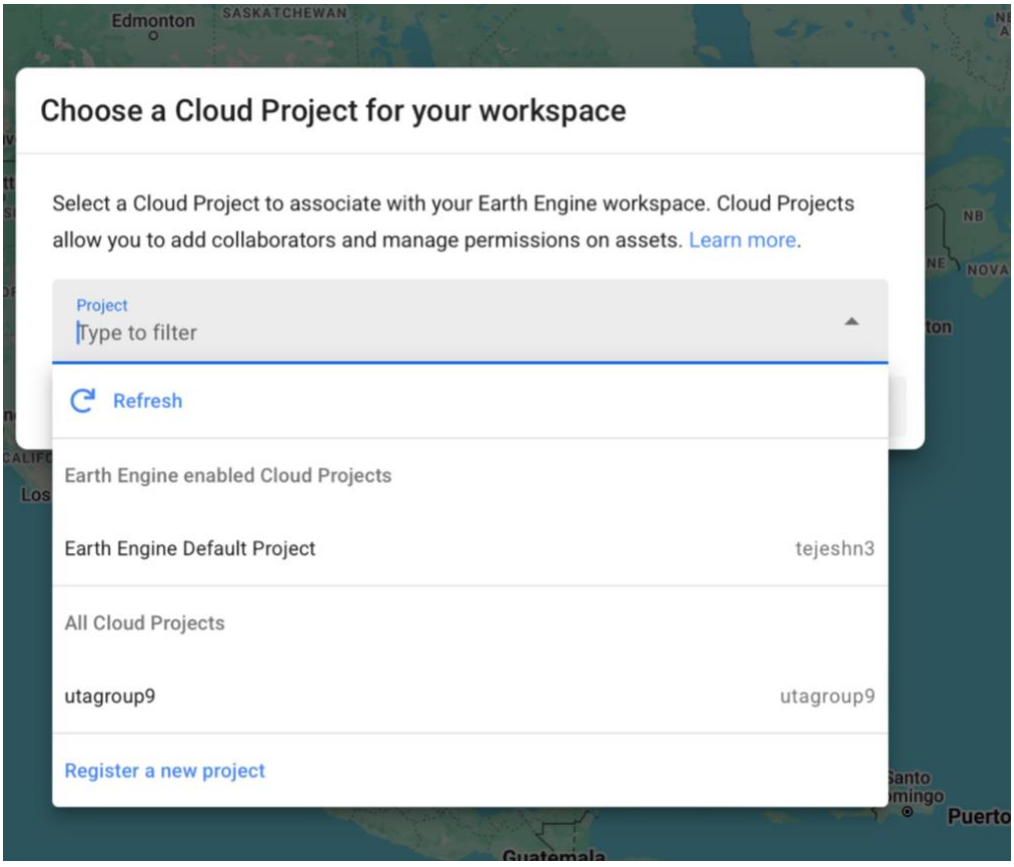


Now select the new created project by clicking on “Select a Project”



The project is now ready to use!

Make your way back to the code editor!



You will now see it listed under projects to use. “utagroup9” in this case is the key which will later be used in the python code.

Phase 2: GEE – Colab – Setup

The infrastructure was set up in Google Colab to provide a cloud-based, scalable environment for working with satellite data using Google Earth Engine (GEE). The goal was to streamline authentication, data access, and geospatial processing without the need for local installations—making the workflow accessible, modular, and easy to replicate.

1. **Setting Up Google Earth Engine (GEE) in Google Colab**

   Google Colab was used as the development environment to eliminate the need for local installations and enable cloud-based access.
   The notebook was connected to the user’s Google Drive to seamlessly access input files such as Excel sheets.

   Import the earthengine-api and geemap.

2. **Authentication and Initialization**

   • Google Earth Engine (GEE) was authenticated through a secure login process within the notebook.
   • The environment was initialized with project-specific credentials to enable access to Earth Engine datasets and tools.

```
3. drive.mount('/content/drive') <- mount your drive
4. ee.Authenticate()
5. ee.Initialize(project='tejeshn3') <- where the project name goes. (make sure collab is logged in using same email in which the project is stored in GEE editor)
6. file_path = '/content/drive/MyDrive/GEE/Modified_Sorghum_BlockReport_1975-1980.xlsx' <- Make sure to upload required dataset in google drive.
7. df = pd.read_excel(file_path)
```

8. **Data Loading and Preparation**

• Input data, such as agricultural yield reports, were uploaded to Google Drive and read into Colab using Python libraries.
• The setup allowed efficient data handling and ensured all required resources were available for visualization.